BACHELOR'S THESIS

Genomic Analysis of Glioblastomas and DIPG: Identification of Potential Therapeutic Targets

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PROJECT DETAILS

- ***Role*** → Biomedical Data Scientist
- ***Supervision & Setting*** → *Universidad Francisco de Vitoria (UFV Madrid, Escuela Politécnica Superior) in direct collaboration with the* ***Brain Tumor Laboratory (Fundación Vithas)***
- ***Technical Stack*** →
  - **Core Programming & Data Engineering:** Python (pandas, numpy, os, glob, re, time)
  - **Statistical Modeling & Hypothesis Testing:** scipy.stats, Pearson Correlation
  - **Machine Learning & Unsupervised Mining:** scikit-learn (StandardScaler, PCA, MDS, LocallyLinearEmbedding, KMeans, DBSCAN, AgglomerativeClustering, NearestNeighbors), umap-learn
  - **Association Rule Mining & Networks:** mlxtend (TransactionEncoder, apriori, association_rules)
  - **Genomic Databases & Bioinformatic Repositories:** COSMIC (Catalogue of Somatic Mutations in Cancer), GeneCards, Gene Ontology (GO), NCBI Genome Data Viewer
  - **Visualization & Reporting:** matplotlib, seaborn, plotly.express, plotly.graph_objects, Microsoft Power BI
- ***Core Domains*** →
  - Whole Exome Sequencing (WES) data processing and filtering
  - Single Nucleotide Variant (SNV) and Copy Number Variant (CNV) analysis
  - Inter-patient and intratumoral heterogeneity modeling
  - Allelic clonality estimation via Variant Allele Frequency (VAF) dynamics
  - Dimensionality reduction and unsupervised clustering validation
  - Data mining for recurrent mutation patterns (Apriori association rules and co-occurrence)
  - Genomic instability assessment via Tumor Mutational Burden (TMB)
- ***Project overview***
  - Designed and built an end-to-end pipeline in Python to analyze Whole Exome Sequencing (WES) data from lethal adult and pediatric brain tumors. Combines statistical hypothesis testing, distance-preserving unsupervised learning, and association rule mining to identify candidate genes and cooperating multi-mutation networks for combination therapies.

CLINICAL CONTEXT & PROBLEM STATEMENT

- **Adult Glioblastoma (GBM):** The most common malignant brain tumor in adults, representing nearly a quarter of all primary brain tumors. Patients survive an average of only 12 to 16 months after diagnosis. Surgery cannot cure it because the tumor cells spread deep into healthy brain tissue like roots, making complete removal impossible. The surviving cells quickly become resistant to standard chemotherapy and radiation.

- **Pediatric DIPG:** A devastating brain tumor that mainly strikes children between 5 and 10 years old, with an average survival of less than a single year. It grows inside the brainstem, the vital structure that controls essential life functions such as breathing, heart rate, and swallowing. Because cutting into this area would be fatal, surgery is impossible. Furthermore, the brain's natural protective shield (the blood-brain barrier) remains tightly sealed in children, blocking most chemotherapy drugs from reaching the tumor cells.

- **The bottleneck:** Standard oncology protocols rely on single-target treatments that consistently fail because brain tumors exhibit extreme **intratumoral heterogeneity,** meaning that different physical regions of the same tumor contain distinct mutation profiles. If a drug targets only one alteration, non-responsive sub-clones survive, expand, and drive fatal recurrence. To find better treatments, tumors cannot be treated as static, isolated gene lists. Methods to identify which mutations occur early versus late, which ones are unique to children versus adults, and which genes work together to keep the cancer alive are needed.

The objective

Main goal

Develop an automated data pipeline in Python that analyzes patient Whole Exome Sequencing (WES) data to discover new genes that could be targeted, providing a data-driven strategy to overcome treatment resistance in adult Glioblastoma and pediatric DIPG.

***Key Steps**:*

1. *Process and clean high-dimensional genomic data*
2. *Isolate early clonal drivers via statistical modeling*
3. *Identify natural patient subtypes*
4. *Discover deterministic pairs of mutations that consistently co-occur or avoid each other.*

DATA PREPROCESSING + TMB ANALYSIS

Dataset scale → 8 patient-derived cell models (6 adult Glioblastomas, 2 pediatric DIPGs) characterized across 286 clinical, genomic, and structural variables, including multi-region biopsies from the same patient.

Data cleaning:

- Filtered the data in order to be aligned with NIH UDP guidelines and eliminate false-positive sequencing artifacts common in repetitive genomic regions.
- Grouped 55 overly granular variant labels into 16 clean biological functional classes.

Tumor Mutational Burden (TMB) analysis

To quantify the structural genomic disruption, the Tumor Mutational Burden (TMB) measures the overall level of genomic instability within a tumor. Rather than looking only at individual point mutations, this analysis calculates what proportion of each chromosome has undergone large-scale structural changes, specifically Copy Number Variations (CNVs) such as multi-gene deletions or duplications. In cancer research, this is critical because tumors with widespread chromosomal instability tend to be far more aggressive, evolve faster, and exhibit higher resistance to standard therapies.

To compute the exact percentage of each chromosome physically altered by CNVs, the following formula was used:

$$\text{Chromosomal disruption (\%)} = \frac{\text{Total unique base pairs altered}}{\text{Total physical length of chromosome}} \times 100$$

This required summing the physical lengths (in base pairs) of all affected genomic regions without counting any physical stretch of DNA more than once.

The problem faced due to structural overlap bug

- **The cause:** When a structural break occurs in cancer DNA, it snaps off a single continuous stretch of the chromosome, removing all genes located along that stretch. Because medical databases are organized by gene name to let clinicians look up specific targets, the database created a separate row for every single gene caught in that break, attaching the full length of the entire event to each individual entry.

- **What failed:** Summing the lengths row by row meant that a single physical deletion spanning 15 genes was counted 15 separate times. This artificial multiplication caused the calculated damage to exceed 100% of the chromosome, which is impossible.

- **Solution:** I built an interval-merging algorithm that operated directly on physical base-pair coordinates rather than gene labels. The script mapped the start and end positions along each chromosome, detected overlapping or nested ranges, and merged them into single, continuous blocks so each damaged section of DNA was counted exactly once. This way, the algorithm eliminated all duplicate counts, replacing impossible numbers with accurate chromosomal disruption metrics between 13% and 31%.

MATHEMATICAL MODELING & METHODOLOGY

Clonality tracking (VAF) → Isolating trunk mutations

- **The idea:** Tumors grow like trees. Early "trunk" mutations are present in every cell, while late "branch" mutations only appear in small parts. Targeting a branch leaves the rest of the tumor alive, so effective therapies must hit the trunk.
- **Methodology:** I tracked **Variant Allele Frequency (VAF)**, which is the percentage of sequencing reads carrying a given mutation. High VAF alterations were flagged as clonal drivers, filtering out late stage noise.

Statistical testing → Prove pediatric tumors are unique

- **The idea:** Historically, children with rare DIPG brain tumors received similar treatments as adults with glioblastomas. We needed to test whether their foundational driver genes were truly different.
- **Methodology:** Because the cohorts had unequal sample sizes and differing variances, standard t-tests were invalid. I implemented **Welch's t-tests**, proving that pediatric tumors are dominated by unique trunk drivers (*ABCA4*, *MUC19*) that do not drive adult tumors, demonstrating mathematically that pediatric patients need their own drug targets.

Dimensionality reduction

- **The idea:** Each patient had 286 genetic variables. To visualize patterns and group similar patients, high-dimensional data must be compressed into 2D/3D without distorting biological relationships.
- **Methodology:** While non-linear techniques like UMAP and t-SNE are standard in bioinformatics for producing visually distinct clusters, they frequently distort global geometry, artificially pulling distant samples together or pushing similar ones apart. To prevent drawing false clinical conclusions from visual artifacts, four techniques were compared:
    - *Linear*: Principal Component Analysis (PCA) and Metric Multidimensional Scaling (MDS).
    - *Non-Linear*: Uniform Manifold Approximation and Projection (UMAP) and Locally Linear Embedding (LLE).
- PCA proved superior, preserving true pairwise geometric distances with 96.12% Fidelity. MDS performed moderately. UMAP and LLE introduced severe geometric distortions. PCA was therefore selected as the best method here.

Unsupervised patient clustering & multi-region validation

To see how the eight patient tumor models naturally grouped together, three standard clustering methods were tested: K-Means, Hierarchical Clustering, and DBSCAN. Instead of creating arbitrary groups, all three algorithms consistently agreed on three clear biological patterns:

- **One unique patient isolated on its own:** Every single algorithm completely separated one adult patient from the rest of the cohort. This tumor had an extreme mutational profile with an exceptionally high number of DNA alterations. DBSCAN automatically flagged it as an isolated outlier, and K-Means placed it in its own individual cluster, confirming it was biologically distinct from typical glioblastomas.
- **Split between kids and adults:** The algorithms grouped the two pediatric DIPG tumors together and separated them from the adult tumors. The models did this purely based on the genetic data without being told patient ages in advance, showing that childhood and adult brain tumors form fundamentally different diseases.
- **Consistency across the same patient:** Two separate biopsies taken from different physical regions of the same patient's tumor clustered tightly together across all algorithms. This confirmed that the analysis was capturing the true biology of the patient rather than random noise from where the biopsy was taken.
- **The typical adult group:** The remaining adult glioblastoma models clustered together, sharing common structural traits like widespread chromosomal deletions and the loss of key protective genes like *PTEN*.

DATA MINING + RESULTS

Data mining → Apriori algorithm (Association rules)

- **Methodology:** Treating each patient's mutation profile like a transactional record, the Apriori algorithm was applied to uncover cooperating mutation networks that drive drug resistance. This identified 148 consistent patterns, revealing that the tumors rarely rely on a single genetic fault. Instead, they systematically work in pairs: the tumor shuts down the body's natural safety mechanisms (genes like *PTEN* or *TP53* that normally prevent abnormal cells from dividing) while simultaneously turning on genes that accelerate aggressive cell growth (like *BRAF*). Pinpointing these specific gene partnerships gave the oncology team clear targets to design drug treatments that attack both vulnerabilities at the same time.

Visualize results on Power BI

To translate these findings into an actionable tool that doesn't involve code and that clinicians can easily use, an interactive Power BI dashboard was implemented for the hospital's brain tumor laboratory. The dashboard turns the complex results into a simple, visual search tool. This way, clinicians can select any gene of interest to immediately see its cooperating mutation partners, visualize where structural damage occurred along the chromosome, and export gene lists to guide drug combination experiments in the lab.

CONCLUSIONS

- **Mathematical proof for dedicated pediatric therapies:** Statistically demonstrated that pediatric DIPG is driven by unique foundational trunk mutations (*ABCA4*, *MUC19*) absent in adult glioblastomas, mathematically proving that children require dedicated targeted drugs rather than adapted adult protocols.
- **Actionable targets for combination therapies:** Identified 148 deterministic mutation partnerships that tumors use to evade single drug treatments, providing concrete gene pairs for multi-target therapies. It specifically paired tumor suppressor loss (*PTEN*, *TP53*) with growth pathway activation (*BRAF*).
- **Genomic processing:** Engineered an automated Python pipeline that resolved the structural overlap bug, reducing genomic processing time from days of manual inspection to minutes while guaranteeing coordinate level accuracy.
- **Translational clinical adoption:** Deployed an interactive Power BI tool directly to the Brain Tumor Laboratory at Fundación Vithas, enabling non-coding oncologists to query candidate gene networks and design in vitro validation experiments.
- **Scientific contribution**: Co-authored the research manuscript *"Potential Oncogenic Role of PRSS1 Identified by Whole Exome Sequencing in Glioma Primary Cell Lines",* submitted for peer-reviewed publication.
- **Academic recognition**: Awarded Highest Honors (Matrícula de Honor) for the Bachelor's Thesis in Mathematical Engineering at Universidad Francisco de Vitoria.